

PROJECT PROPOSAL

Programming Fundamentals (Lab)

~ Kaun Banega Crorepati ~

Student Information

Project Name: Kaun Banega Crorepati (KBC) — Quiz Game

Student: Ahmad

Registration #:

Department:

University: University of Engineering and Technology, Lahore

Semester:

Course Subject: Programming Fundamentals

1. Introduction

Kaun Banega Crorepati (KBC) Quiz Game is a Python-based command-line game inspired by the world-famous television quiz show. The player is asked 15 multiple-choice questions of increasing difficulty, each with a prize money value that doubles with every correct answer. The player can use three lifelines — 50:50 (removes two wrong options), Audience Poll (shows a percentage hint), and Phone a Friend (gives a random suggestion) — once each during the game. The game ends when the player answers all 15 questions, answers incorrectly, or chooses to quit and take their winnings. All questions are loaded from a file, making it easy to expand the question bank.

2. Problem Statement

Quiz-based learning tools are highly effective for students, yet most beginner Python projects lack engagement and real-world inspiration. There is a need for a Python CLI project that goes beyond a basic quiz by incorporating progressive difficulty, prize money tracking, lifeline mechanics, and a dramatic game-show atmosphere — all while remaining buildable using only fundamental Python concepts. The KBC Quiz Game addresses this by delivering a feature-rich, exciting, and educationally valuable project that any first-semester student can understand and build.

3. Objectives

- Load 15 multiple-choice questions from a questions file (questions.txt or questions.json).
- Present questions one by one in increasing difficulty with 4 answer options (A, B, C, D).
- Display the prize money ladder showing the player's current winnings after each correct answer.

- Implement the 50:50 lifeline — randomly removes 2 of the 3 wrong answer options.
- Implement the Audience Poll lifeline — displays a fake percentage hint favoring the correct answer.
- Implement the Phone a Friend lifeline — displays a hint message suggesting an answer.
- Allow the player to quit at any time and walk away with their current guaranteed prize.
- End the game with a loss message (wrong answer), win message (all 15 correct), or quit summary.
- Display a final scoreboard showing questions answered, lifelines used, and prize money won.

4. Scope of Project

The project is a single-file Python CLI application. Its scope covers:

- Question Bank: 15 questions stored in questions.json with question, options A-D, and correct answer.
- Prize Ladder: 15 levels — Rs. 1,000 → Rs. 10,000 → Rs. 1,00,000 → Rs. 1,00,00,000 (1 Crore).
- Safe Havens: Questions 5 and 10 are guaranteed prizes (player keeps this even if they lose after).
- Lifelines: 50:50, Audience Poll, Phone a Friend — each usable once per game.
- Quit Option: Player can type 'quit' at any question to take current winnings.
- Wrong Answer: Game ends, player wins only up to the last safe haven.

The project does not include a GUI, real networking, or OOP.

5. Technologies Used

Technology	Purpose
Python 3.x	Core programming language
json (stdlib)	Loading the question bank from questions.json file
random (stdlib)	50:50 lifeline — randomly selecting 2 wrong options to remove
time (stdlib)	Adding dramatic pauses between question reveal and answer check
os (stdlib)	Clearing the terminal screen between questions for clean display

6. Python Topics Used

Python Topic	How It Is Used in the Project
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Variables & Data Types	Prize money (int), player answer (string), lifeline flags (bool)
Lists	Storing answer options, prize ladder amounts, and available lifelines
Dictionaries	Each question stored as a dict: {question, A, B, C, D, answer}
Loops (for / while)	Main game loop iterates over 15 questions; input loops for validation
Functions	Separate functions for ask_question, use_5050, audience_poll, phone_friend, show_ladder
File Handling (JSON)	Loading all 15 questions from questions.json at game startup
Conditional Statements	Win/lose/quit detection; lifeline availability checks; safe haven logic
Boolean Variables	Tracking which lifelines have been used (used_5050, used_poll, used_phone)
String Methods	.upper().strip() to normalize player's answer input (a/A/b/B etc.)
random module	random.sample() for 50:50; random.choice() for Phone a Friend hint

7. Expected Output

On startup:

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KAUN BANEGA CROREPATI - KBC GAME
Welcome, Contestant!

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Lifelines available: 50:50 | Audience Poll | Phone a Friend

Sample question screen:

Question 3 of 15 Prize: Rs. 3,200

Which planet is known as the Red Planet?

A. Venus B. Mars
C. Jupiter D. Saturn

Lifelines: [50:50] [Poll] [Phone]

Your answer (A/B/C/D) or 'lifeline' or 'quit': B

✔ **Correct! You have won Rs. 3,200!**

After using 50:50 lifeline:

50:50 used! Remaining options:
A. Venus B. Mars

Game over on wrong answer:

✘ Wrong! The correct answer was B. Mars
You reached safe haven: Rs. 1,000
You walk away with: Rs. 1,000

Final scoreboard on winning all 15:

☐ CONGRATULATIONS! YOU ARE A CROREPATI!
Questions answered : 15/15
Lifelines used : 50:50, Phone a Friend
Prize won : Rs. 1,00,00,000 (1 CRORE!)

8. Benefits of Project

- Highly engaging and recognizable — everyone knows KBC, making the viva demo exciting.
- Progressive difficulty and prize ladder teach logical game state management.
- Three distinct lifeline mechanics reinforce conditional logic and boolean tracking.
- JSON-based question bank teaches structured file handling and easy content expansion.
- Safe haven logic introduces the concept of checkpoints — a real game design pattern.
- One of the most impressive-looking CLI projects for a beginner's portfolio.

9. Future Enhancements

- Add a question difficulty system — Easy (Q1-5), Medium (Q6-10), Hard (Q11-15) — with separate question pools.
- Implement a high score leaderboard saved to a file with player name and prize won.
- Add a timer per question (e.g. 30 seconds) using the time module for added pressure.
- Build a multiplayer mode where two players compete in alternating rounds.
- Create a GUI version using Tkinter with a prize ladder displayed on the side panel.
- Add background music and sound effects using the playsound library for a TV show atmosphere.

10. Conclusion

The Kaun Banega Crorepati Quiz Game is one of the most impressive and feature-complete beginner Python projects possible within the scope of a Programming Fundamentals lab course. It combines question loading from a JSON file, prize ladder progression, three lifeline mechanics, safe haven checkpoints, and a dramatic game-show atmosphere — all built using only variables, lists, dictionaries, loops, functions, file handling, and the random module. The project is immediately recognizable, genuinely fun to play, and makes for an outstanding viva demonstration. By completing this project, students gain deep practical experience with all core Python fundamentals while producing something they can be truly proud of.